

Code:-

```
package IPProcess;

import java.io.IOException;
import java.util.StringTokenizer;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
import org.apache.hadoop.mapreduce.lib.reduce.IntSumReducer;

public class IPProcess {

    public static class TokenizerMapper extends Mapper<Object, Text, Text, IntWritable>{

        private final static IntWritable one = new IntWritable(1);
        private Text word = new Text();

        public void map(Object key, Text value, Context context) throws IOException,
        InterruptedException{

            //logic starts here
            String[] tokens = value.toString().split(" ");
            String ipAddress = tokens[0];

            word.set(ipAddress);
            context.write(word, one);
        }
    }

    public static class intSumReducer extends Reducer<Text, IntWritable, Text, IntWritable>{

        private IntWritable result = new IntWritable();

        public void reduce(Text key, Iterable<IntWritable> values, Context context)
        throws IOException, InterruptedException{

            int sum = 0;
            for (IntWritable val : values){
```

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        sum +=val.get();
    }
    result.set(sum);
    context.write(key, result);
}

}

public static void main(String[] args) throws Exception {
    Configuration conf = new Configuration();
    Job job = Job.getInstance(conf, "word count");
    job.setJarByClass(IPProcess.class);
    job.setMapperClass(TokenizerMapper.class);
    job.setCombinerClass(IntSumReducer.class);
    job.setReducerClass(IntSumReducer.class);
    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(IntWritable.class);
    FileInputFormat.addInputPath(job, new Path(args[0]));
    FileOutputFormat.setOutputPath(job, new Path(args[1]));
    System.exit(job.waitForCompletion(true) ? 0 : 1);
}

}

```

Output:-

```

[cloudera@quickstart ~]$ pwd
/home/cloudera
[cloudera@quickstart ~]$ ls
cloudera-manager eclipse          lib      Videos
cm_api.py      enterprise-deployment.json Music    workspace
Desktop        express-deployment.json  Pictures
Documents      IPProcess.jar           Public
Downloads      kerberos                Templates
[cloudera@quickstart ~]$ hadoop fs -put Desktop/logfi/access_log_short.txt inputlog.txt
[cloudera@quickstart ~]$ hadoop fs -ls
Found 1 items
-rw-r--r--  1 cloudera cloudera   158551 2026-03-19 05:34 inputlog.txt
[cloudera@quickstart ~]$ hadoop jar IPProcess.jar IPProcess.IPProcess inputlog.txt op60
26/03/19 05:40:13 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
26/03/19 05:40:15 WARN mapreduce.JobSubmitter: Hadoop command-line option parsing not
performed. Implement the Tool interface and execute your application with ToolRunner to
remedy this.
26/03/19 05:40:16 INFO input.FileInputFormat: Total input paths to process : 1
26/03/19 05:40:17 INFO mapreduce.JobSubmitter: number of splits:1

```

26/03/19 05:40:17 INFO mapreduce.JobSubmitter: Submitting tokens for job:
job_1773910366565_0001
26/03/19 05:40:20 INFO impl.YarnClientImpl: Submitted application
application_1773910366565_0001
26/03/19 05:40:20 INFO mapreduce.Job: The url to track the job:
http://quickstart.cloudera:8088/proxy/application_1773910366565_0001/
26/03/19 05:40:20 INFO mapreduce.Job: Running job: job_1773910366565_0001
26/03/19 05:41:16 INFO mapreduce.Job: Job job_1773910366565_0001 running in uber mode
: false
26/03/19 05:41:16 INFO mapreduce.Job: map 0% reduce 0%
26/03/19 05:41:41 INFO mapreduce.Job: map 100% reduce 0%
26/03/19 05:42:05 INFO mapreduce.Job: map 100% reduce 100%
26/03/19 05:42:11 INFO mapreduce.Job: Job job_1773910366565_0001 completed
successfully
26/03/19 05:42:13 INFO mapreduce.Job: Counters: 49
 File System Counters
 FILE: Number of bytes read=4479
 FILE: Number of bytes written=229727
 FILE: Number of read operations=0
 FILE: Number of large read operations=0
 FILE: Number of write operations=0
 HDFS: Number of bytes read=158674
 HDFS: Number of bytes written=3611
 HDFS: Number of read operations=6
 HDFS: Number of large read operations=0
 HDFS: Number of write operations=2
 Job Counters
 Launched map tasks=1
 Launched reduce tasks=1
 Data-local map tasks=1
 Total time spent by all maps in occupied slots (ms)=20619
 Total time spent by all reduces in occupied slots (ms)=22779
 Total time spent by all map tasks (ms)=20619
 Total time spent by all reduce tasks (ms)=22779
 Total vcore-seconds taken by all map tasks=20619
 Total vcore-seconds taken by all reduce tasks=22779
 Total megabyte-seconds taken by all map tasks=21113856
 Total megabyte-seconds taken by all reduce tasks=23325696
 Map-Reduce Framework
 Map input records=1294
 Map output records=1294
 Map output bytes=22883
 Map output materialized bytes=4479
 Input split bytes=123
 Combine input records=1294
 Combine output records=227

Reduce input groups=227
Reduce shuffle bytes=4479
Reduce input records=227
Reduce output records=227
Spilled Records=454
Shuffled Maps =1
Failed Shuffles=0
Merged Map outputs=1
GC time elapsed (ms)=75
CPU time spent (ms)=3340
Physical memory (bytes) snapshot=337047552
Virtual memory (bytes) snapshot=3007135744
Total committed heap usage (bytes)=226365440

Shuffle Errors

BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0

File Input Format Counters

Bytes Read=158551

File Output Format Counters

Bytes Written=3611

[cloudera@quickstart ~]\$ hadoop fs -cat dir60/part-r-00000

cat: `dir60/part-r-00000': No such file or directory

[cloudera@quickstart ~]\$ hadoop fs -cat op60/part-r-00000

10.1.1.236	7
10.1.181.142	14
10.1.232.31	5
10.10.55.142	14
10.102.101.66	1
10.103.184.104	1
10.103.190.81	53
10.103.63.29	1
10.104.73.51	1
10.105.160.183	1
10.108.91.151	1
10.109.21.76	1
10.11.131.40	1
10.111.71.20	8
10.112.227.184	6
10.114.74.30	1
10.115.118.78	1
10.117.224.230	1
10.117.76.22	12

10.118.19.97	1	
10.118.250.30	7	
10.119.117.132		23
10.119.33.245	1	
10.119.74.120	1	
10.12.113.198	2	
10.12.219.30	1	
10.120.165.113		1
10.120.207.127		4
10.123.124.47	1	
10.123.35.235	1	
10.124.148.99	1	
10.124.155.234		1
10.126.161.13	7	
10.127.162.239		1
10.128.11.75	10	
10.13.42.232	1	
10.130.195.163		8
10.130.70.80	1	
10.131.163.73	1	
10.131.209.116		5
10.132.19.125	2	
10.133.222.184		12
10.134.110.196		13
10.134.242.87	1	
10.136.84.60	5	
10.14.2.86	8	
10.14.4.151	2	
10.140.139.116		1
10.140.141.1	9	
10.140.67.116	1	
10.141.221.57	5	
10.142.203.173		7
10.143.126.177		32
10.144.147.8	1	
10.15.208.56	1	
10.15.23.44	13	
10.150.212.239		14
10.150.227.16	1	
10.150.24.40	13	
10.152.195.138		8
10.153.23.63	2	
10.153.239.5	25	
10.155.95.124	9	
10.156.152.9	1	
10.157.176.158		1

10.164.130.155	1
10.164.49.105	8
10.164.95.122	10
10.165.106.173	14
10.167.1.145	19
10.169.158.88	1
10.170.178.53	1
10.171.104.4	1
10.172.169.53	18
10.174.246.84	3
10.175.149.65	1
10.175.204.125	15
10.177.216.164	6
10.179.107.170	2
10.181.38.207	13
10.181.87.221	1
10.185.152.140	1
10.186.56.126	16
10.186.56.183	1
10.187.129.140	6
10.187.177.220	1
10.187.212.83	1
10.187.28.68	1
10.19.226.186	2
10.190.174.142	10
10.190.41.42	5
10.191.172.11	1
10.193.116.91	1
10.194.174.4	7
10.198.138.192	1
10.199.103.248	2
10.199.189.15	1
10.2.202.135	1
10.200.184.212	1
10.200.237.222	1
10.200.9.128	2
10.203.194.139	10
10.205.72.238	2
10.206.108.96	2
10.206.175.236	1
10.206.73.206	7
10.207.190.45	17
10.208.38.46	1
10.208.49.216	4
10.209.18.39	9
10.209.54.187	3

10.211.47.159	10
10.212.122.173	1
10.213.181.38	7
10.214.35.48	1
10.215.222.114	1
10.216.113.172	48
10.216.134.214	1
10.216.227.195	16
10.217.151.145	10
10.217.32.16	1
10.218.16.176	8
10.22.108.103	4
10.220.112.1	34
10.221.40.89	5
10.221.62.23	13
10.222.246.34	1
10.223.157.186	10
10.225.137.152	1
10.225.234.46	1
10.226.130.133	1
10.229.60.23	1
10.230.191.135	6
10.231.55.231	1
10.234.15.156	1
10.236.231.63	1
10.238.230.235	1
10.239.100.52	1
10.239.52.68	4
10.24.150.4	5
10.24.67.131	13
10.240.144.183	15
10.240.170.50	1
10.241.107.75	1
10.241.9.187	1
10.243.51.109	5
10.244.166.195	5
10.245.208.15	20
10.246.151.162	3
10.247.111.104	9
10.247.175.65	1
10.247.229.13	1
10.248.24.219	1
10.248.36.117	3
10.249.130.132	3
10.25.132.238	2
10.25.44.247	6

10.250.166.232	1
10.27.134.23	1
10.30.164.32	1
10.30.47.170	8
10.31.225.14	7
10.32.138.48	11
10.32.247.175	4
10.32.55.216	12
10.33.181.9	8
10.34.233.107	1
10.36.200.176	1
10.39.45.70	2
10.39.94.109	4
10.4.59.153	1
10.4.79.47	15
10.41.170.233	9
10.41.40.17	1
10.42.208.60	1
10.43.81.13	1
10.46.190.95	10
10.48.81.158	5
10.5.132.217	1
10.5.148.29	1
10.50.226.223	9
10.50.41.216	3
10.52.161.126	1
10.53.58.58	1
10.54.242.54	10
10.54.49.229	1
10.56.48.40	16
10.59.42.194	11
10.6.238.124	6
10.61.147.24	1
10.61.161.218	1
10.61.23.77	8
10.61.232.147	3
10.62.78.165	2
10.63.233.249	7
10.64.224.191	13
10.66.208.82	2
10.69.20.85	26
10.70.105.238	1
10.70.238.46	6
10.72.137.86	6
10.72.208.27	1
10.73.134.9	4

```
10.73.238.200 1
10.73.60.200 1
10.73.64.91 1
10.74.218.123 1
10.75.116.199 1
10.76.143.30 1
10.76.68.178 16
10.78.95.24 8
10.80.10.131 10
10.80.215.116 17
10.81.134.180 1
10.82.30.199 63
10.82.64.235 1
10.84.236.242 1
10.87.209.46 1
10.87.88.214 1
10.88.204.177 1
10.89.178.62 1
10.89.244.42 1
10.94.196.42 1
10.95.136.211 4
10.95.232.88 1
10.98.156.141 1
10.99.228.224 1
[cloudera@quickstart ~]$
```